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Full Length Article

# Cost observability as a security control in multi-cloud environments based on SOC 2 security standard

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ABSTRACT

The problem of ensuring cost transparency and proactive budget control in multi-cloud environments is becoming increasingly relevant for modern IT infrastructures. As organizations scale their use of heterogeneous cloud services, they face challenges related to fragmented billing systems, inconsistent cost metrics, and delayed anomaly detection. This study frames cost observability not merely as a financial function, but as an integral component of the organization's security posture, aligned with the SOC 2 framework. The novelty of this research lies in the integration of cost monitoring tools — specifically Splunk, Cherwell, and JSON-based cloud Application Programming Interfaces (APIs) — with operational and security processes, enabling real-time detection of budget deviations, automated incident escalation, and cost-based policy enforcement. This study presents a proposed future architecture that implements a unified cost observability layer across heterogeneous billing systems in multi-cloud environments. The architecture transforms provider-specific formats — including AWS Cost Explorer JSON exports, Azure Cost Management APIs, and GCP Billing BigQuery exports — into standardized cost events. These normalized streams create a single temporal view of expenditures against unified budget thresholds, while generating consolidated financial telemetry that enables cross-provider anomaly detection and correlation. By reframing cost data as actionable observability signals, the approach advances beyond fragmented dashboards toward a centralized, audit-ready control layer that supports compliance, incident response, and financial governance. The system further incorporates role-based access logic, escalation thresholds, and forecasting models, creating a cost governance layer with direct implications for FinOps, DevSecOps, and compliance teams.

## 1. Introduction and problem statement

Modern IT organizations increasingly face challenges in maintaining cost transparency and consistent financial governance across multiple cloud providers. Differences in billing models, data structures, and reporting mechanisms lead to fragmented visibility, delayed anomaly detection, and limited auditability.

This study presents a proposed unified cost observability architecture that consolidates financial and operational data into a single, queryable view. The system transforms disparate billing streams from AWS, Azure, and GCP into standardized cost events, enabling unified analysis and anomaly correlation across providers. This approach shifts the paradigm from isolated dashboards to a centralized observability layer that integrates budgeting, monitoring, and compliance workflows under a common framework aligned with SOC 2 Type II. It provides role-based accountability, escalation logic, and forecasting mechanisms that extend financial observability into the organization's broader security posture

Such integration is especially relevant in the context of compliance with external information security standards, particularly SOC (Service Organization Control) 2. These standard mandates the implementation of control procedures within categories such as security, availability, and processing integrity (TSC). In this regard, cost monitoring is viewed not only as a financial optimization tool but also as part of the operational security model, where financial anomalies can serve as indicators of risk, and the system itself provides evidence for audits.

## 2. Research objective

The primary goal of this research is to develop and justify an architectural solution for cost monitoring in a multi-cloud environment that not only implements Financial Operations (FinOps) principles but also

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ensures compliance with information security criteria according to SOC 2 requirements. This approach allows for the integration of technical, financial, and audit control within a unified system aimed at enhancing transparency, manageability, and trust in the cloud infrastructure.

Using a visual example from real-world practice, the paper analyzes the process of building such a system, its integration components, escalation logic, and the roles of stakeholders. Additionally, a review of current academic research is presented, focusing on financial monitoring in cloud environments, particularly in the context of FinOps and Cloud Cost Governance.

### 2.1. Notation and terminology

This subsection defines the acronyms, roles, and mathematical symbols used throughout the paper to ensure clarity and consistency.

**Acronyms:**

- SOC 2 - Service Organization Control 2
- TSC - Trust Services Criteria
- FinOps - Financial Operations
- CMDB - Configuration Management Database
- SIEM - Security Information and Event Management
- ITSM - IT Service Management
- CSA - Cloud Service Allocation

**Role Designations:**

- Budget Owner — individual responsible for project-level spending
- CSA Owner — Cloud Service Allocation owner with cross-project authority
- Project Manager — technical lead for workload management

**Mathematical Symbols:**

- $C(t)$: actual accumulated cost at time $t$
- $B$: approved budget
- $\theta_k$: predefined cost threshold (e.g., 0.7, 0.9, 1.0)
- $\alpha$: cost growth rate
- $\beta$: initial accumulated cost at the beginning of the billing period
- $t_k^*$: estimated time when threshold $\theta_k$is reached

## 3. Methodology

To achieve the stated objective, this study applies a practice-oriented architectural research methodology, grounded in the analysis of real-world scenarios and operational data. The methodological approach begins with the development of an applied system design that integrates well-established technological components. Specifically, the solution utilizes Splunk for data aggregation and cost analytics, Cherwell as the platform for incident and request management, and native billing APIs from major cloud providers (AWS, Azure, GCP) to collect cost-related data in near real time.

The research also incorporates scenario-based analysis, which explores practical use cases related to financial threshold violations, automated alerting, and escalation logic distributed across multiple organizational roles. Particular attention is paid to the configuration of budget consumption thresholds, timing of system responses, and role-based notification mechanisms, reflecting real operational practices.

To assess the alignment of the proposed system with recognized security standards, the implemented architecture is mapped against the SOC 2 TSC. This mapping identifies how specific cost-related events, such as unexpected budget overruns or deviations from forecasted consumption, can serve as operational triggers within security and compliance workflows, particularly those related to processing integrity, availability, and monitoring activities.

The methodological framework also includes a review of relevant academic and industry literature, focusing on current approaches to cloud cost governance, the evolution of FinOps practices, and compliance mechanisms in distributed cloud environments. This literature foundation supports the theoretical justification for treating cost observability as a form of operational control.

Finally, the effectiveness of the proposed model is validated through retrospective analysis of historical billing data. This stage involves examining changes in key indicators—such as budget adherence, escalation timing, and financial transparency—following the system's implementation. The findings from this empirical assessment inform the conclusions regarding the practical impact and replicability of the approach in enterprise-scale multi-cloud infrastructures.

Confirmation of this hypothesis allows multi-cloud infrastructure to be considered not just as a technical solution but as a managed environment in which cost transparency and control become critical factors for sustainable development, financial stability, and compliance with external audit standards.

## 4. Literature review

The issue of cost monitoring and optimization in multi-cloud environments is actively explored in contemporary scientific literature, particularly in the context of increasing complexity in managing financial resources across heterogeneous cloud platforms. In recent years, research has shifted from purely technical aspects of infrastructure management toward integrated approaches that combine computing resources, cost analytics, and financial management practices.

In the work by Alexander et al. (2020) a cost-aware orchestration model is proposed, which integrates the TOSCA standard with a custom cost assessment and forecasting policy. The authors argue that accounting for future costs during orchestration not only enables flexible application scaling but also helps keep expenses within acceptable budget limits. This represents a significant step toward building dynamic, self-regulating financial control systems in multi-cloud environments.

National research, such as the article by Shokotko et al. (2024), examines a three-phase approach to cloud cost management, which includes understanding cost drivers, controlling budgets, and optimizing expenditures. The authors emphasize the importance of implementing FinOps practices to improve financial transparency and recommend using built-in tools from cloud providers for daily monitoring.

The challenge of selecting an optimal cloud provider based on query execution costs is addressed in the study by Wojtowicz et al. (2022). The proposed model allows for the formation of an optimal query plan based on cost and performance, which is highly relevant for companies dealing with large volumes of data in multi-cloud environments. This approach highlights the need for deeper integration of cost monitoring at the level of analytical queries and data management systems.

The study by Li et al. (2022) focuses on the practical implementation of the SmartCMP platform, which automates cost monitoring and provides centralized policy management for optimization in multi-cloud environments. The platform enables budget setting, anomaly detection, and the triggering of actions to reduce overspending, demonstrating the practical effectiveness of FinOps approaches.

Special attention should be given to the study by Thumala and Pillai (2024), which focuses on cost optimization during cloud migration. The authors describe rightsizing methodologies, management of reserved instances, optimization of inter-cloud network interactions, and evaluate the effectiveness of different financial strategies depending on workload type. This confirms the necessity of a multi-level approach to cost management that includes both engineering and financial solutions.

In summary, the scientific literature confirms that effective cost management in a multi-cloud environment requires:

- The use of dynamic cost modeling approaches,

- Implementation of integrated platforms for collecting and analyzing financial data,
- Application of FinOps and governance practices to align IT and financial teams,
- Cost forecasting and timely anomaly response via automated alerts and incident management

In today's digital landscape—where cloud technologies form the foundation for scalable, flexible, and highly available IT solutions—cost monitoring takes on strategic importance. The growing popularity of multi-cloud architectures, which involve simultaneous use of services from various cloud providers, intensifies the need for cost transparency, financial risk forecasting, and effective budget management (Thumala and Pillai, 2024).

The reviewed scientific sources clearly indicate that effective cost monitoring is not just a financial control tool but a key factor in ensuring the stability and predictability of IT services. Specifically, the combination of monitoring tools (Splunk, SmartCMP), automated overspend response, FinOps practices, and integration with CMDB solutions (e.g., Cherwell) enables organizations to make informed management decisions based on real-world data.

Thus, cost monitoring in a multi-cloud environment is evolving from a support function into a critically important component of cloud governance—ensuring a balance between technical efficiency and financial accountability. This highlights the need for further research and the deployment of systems that integrate technical analytics, business logic, and automated response to ensure the sustainable operation of modern IT ecosystems (Konidena, 2024).

## 5. The role of FinOps in cost monitoring in multi-cloud infrastructure

FinOps has emerged as the methodological foundation for transparent and accountable use of cloud resources in multi-cloud environments. Positioned at the intersection of finance, development, and operations, it provides a unified framework where stakeholders share a common understanding of spending volumes, control mechanisms, and cost-optimization practices (Chornii et al., 2025).

In multi-cloud settings, where billing models differ and cost tracking is fragmented, FinOps enables normalization of budget metrics across providers (AWS, Azure, GCP) and alignment of spending with business objectives. Within the proposed architecture, FinOps principles are realized through cost aggregation and budget comparison in Splunk, automated threshold detection, escalation via Cherwell with role-based responsibilities for DevSecOps and SecOps teams, and monthly forecasting supported by Splunk dashboards.

Thus, FinOps is not limited to tools but functions as a conceptual framework linking financial metrics with technical events and organizational responsibilities. Importantly, it also supports SOC 2 criteria — particularly Processing Integrity and Security — by ensuring that cost anomalies are treated as part of the organization's broader control environment (Piskozub and Abibulaiev, 2025)

### 5.1. SOC 2 as security control for cloud spending

In the context of managing multi-cloud infrastructure, more and more organizations aim to ensure not only technical efficiency in resource usage but also compliance with security, transparency, and accountability requirements. One of the most widely adopted frameworks that formalizes these requirements is SOC 2 — a system of controls developed by the American Institute of Certified Public Accountants (AICPA) for assessing the effectiveness of information security in organizations that provide cloud and technology services.

SOC 2 is based on the TSC — principles that define the expected behavior of an organization in the areas of Security, Availability, Processing Integrity, Confidentiality, and Privacy.

SOC 2 reports are conducted under the Statement on Standards for Attestation Engagements No 18 (SSAE-18), which provides the basis for evaluating the design and effectiveness of controls. Two types of SOC 2 reports exist: Type I, which assesses the design of controls at a specific point in time, and Type II, which evaluates the operational effectiveness of those controls over a defined observation period. Since this study addresses cost monitoring as a continuous operational process, the analysis is aligned primarily with SOC 2 Type II. The discussion is focused on the TSC most relevant to financial observability and risk response in multi-cloud environments — specifically Security, Availability, Processing Integrity, and Monitoring Activities. This alignment emphasizes the system's ability to demonstrate control effectiveness over time, rather than adequacy of design alone.

Within this framework, cost monitoring can be operationalized as a SOC 2 Type II control, supporting compliance and risk mitigation in multi-cloud environments (Kamal et al., 2024).

Cost monitoring with threshold-based alerts, incident escalation, and response history can be regarded as a control activity that enables:

- Detection of deviations from expected system behavior, which may result from unauthorized or improper use of resources.
- Verification of proper functioning of IT processes (e.g., automatic shutdown of excessive instances or reaction to budget overruns);
- Event logging for audit purposes, including log capture in SIEM systems or services like Splunk.
- Allocation of roles and responsibilities (CSA, FinOps, DevSecOps) according to access management policies, which is part of SOC 2 accountability requirements.

Integrating cost monitoring with systems like Cherwell, ServiceNow, and Splunk enables the implementation of a multi-tiered response model where each financial event is treated as a potential business or security event. This aligns with the procedural requirements for SOC 2 Type II, where consistency of processes over time is critical (Deineka and Bortnik, 2024). Thus, cost monitoring in a multi-cloud environment can not only optimize financial consumption but also serve as an operational control within the SOC 2 framework. This approach bridges FinOps and information security, creating a foundation for comprehensive compliance and increased trust from clients, partners, and auditors.

Ultimately, incorporating cloud cost monitoring as part of the organization's security strategy contributes to a broader security posture. Financial anomalies often reflect deeper operational or configuration issues—such as misallocated resources, unauthorized deployments, or policy violations—that may not be immediately visible through traditional technical monitoring alone. By treating unexpected spending as a potential risk indicator, organizations gain an additional dimension of situational awareness, which enhances both preventive and detective controls.

In this way, cloud spending oversight becomes a meaningful extension of the organization's security operations. It adds a layer of business-centric visibility to technical environments, strengthens accountability mechanisms, and supports early detection of behaviors that could indicate breaches, misuse, or compliance failures. Therefore, integrating cloud cost monitoring into SOC 2-aligned security frameworks not only satisfies audit expectations but also improves the organization's ability to manage risk proactively and holistically.

### 5.2. Mapping to SOC 2 TSC and risk model

The mapping framework establishes the relationship between system controls and the TSC most pertinent to security risk mitigation. The underlying risk model is adapted from the NIST SP 800–30 "Guide for Conducting Risk Assessments," ensuring a structured and standardized approach to threat identification, likelihood estimation, and impact analysis. Cost monitoring functions are examined in terms of their contribution to reducing specific threat scenarios, where financial

anomalies act as secondary indicators of operational or security events.

For each identified threat scenario, both the probability of occurrence and its potential impact on SOC 2 criteria—particularly Security, Availability, and Processing Integrity—were evaluated. The primary threats considered are:

- **Unauthorized resource provisioning** — indicative of compromised access or bypassed change controls.
- **Untracked or shadow workloads** — leading to uncontrolled expenditures and policy non-compliance.
- **Budget exhaustion** — resulting in potential service interruptions and breaches of availability commitments.
- **Ineffective alert configuration** — delaying detection and response to anomalous activities.
- **Misuse of resources** — generating financial losses and regulatory exposure.

The identified risk scenarios are mapped against the SOC 2 TSC to demonstrate the explicit compliance coverage provided by the implemented controls. This mapping links each relevant criterion to a concrete technical measure within the cost monitoring system and specifies the corresponding risk mitigated. The focus is on criteria directly related to security, availability, and processing integrity, reflecting how financial observability supports both preventive and detective controls in multi-cloud environments.

This mapping illustrates that financial observability functions extend beyond budget governance, forming an integral part of the organization's security control environment. By aligning cost monitoring mechanisms with SOC 2 TSC, the system enables early detection of anomalous activities, enforces role-based accountability, and preserves audit-ready evidence of incident handling. Consequently, cloud spending oversight becomes both a compliance enabler and a proactive risk mitigation tool, strengthening the overall resilience and trustworthiness of multi-cloud operations.

### 5.3. Overview of it solutions for monitoring cloud costs

Managing costs in cloud environments is not just about reducing bills; it's primarily about ensuring transparency, predictability, and financial discipline in conditions of dynamic resource consumption. With the growing complexity of multi-cloud infrastructure, organizations face new challenges: inconsistency in reporting formats among providers, differences in pricing models, delays in data updates, and insufficient integration of financial tools with the technical infrastructure (Sapsai et al., 2025).

In this context, the importance of both a methodologically sound analytical approach and the selection of appropriate IT solutions capable of supporting the full monitoring cycle — from data collection to automated response — is increasing. Building an effective cost control system requires consideration not only of technical aspects of metric collection but also of organizational processes related to budgeting, analytics, and interdepartmental collaboration.

Most cloud service providers — such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP) — offer their own services for cost monitoring and control. These services are designed to provide both basic and advanced tools for cost analysis, budget planning, and alerts when financial thresholds are exceeded (Cenaj et al., 2025).

### 5.4. Built-in cost monitoring services from cloud providers

First-tier cloud providers — AWS, Azure, and GCP — offer their own tools for tracking and controlling spending. Services like AWS Cost Explorer, Azure Cost Management and Google Cloud Billing Reports, provide basic analytics, budget creation, alert configuration for threshold breaches, and report generation.

These tools allow users to visualize data in real time, generate forecasts, and implement an initial level of financial control without relying on external solutions. However, their use is limited to a single provider's infrastructure, which makes comprehensive control in a multi-cloud environment impossible. Therefore, built-in services are often used as a foundation for further integration with independent FinOps-class platforms (Opirskyy et al., 2025).

**AWS: Cost Explorer and AWS Budgets.**

AWS Cost Explorer is a visual tool that allows users to analyze cloud spending through charts and tables. It enables users to:

- View expenses daily, monthly, or for a custom period.
- Group costs by services (EC2, S3, RDS, etc.), tags, and accounts.
- Detect anomalies or unexpected cost increases.
- Analyze spending trends for forecasting.

The AWS Budgets service allows users to set individual budget limits for specific accounts, cloud services, projects, or resources with designated tags. When a defined spending threshold is reached (e.g., 80 % of the set budget), the system automatically generates alerts that can be sent via email or the Amazon SNS messaging service.

These services are integrated into the AWS Billing Dashboard and are free of charge (some advanced analytics features may be paid for with large data volumes) (Banala, 2025).

**Microsoft Azure: Cost Management + Billing.**

Azure Cost Management + Billing is a comprehensive toolkit for monitoring, analyzing, and optimizing costs in the Microsoft Azure cloud environment. The system provides detailed visualization of financial metrics by subscription, resource groups, individual resources, tags, and specific time periods. One of its key features is the automated generation of budget reports with the ability to send them to designated recipients.

This service is particularly convenient for large organizations actively using Office 365 or Power Platform, as it integrates well with other Microsoft tools (Mittal, 2025).

**Google Cloud Platform: Billing reports and budgets.**

Google Cloud Billing Reports is a visual interface designed for analyzing cloud expenditures within the Google Cloud ecosystem. The tool enables detailed analytics by project, account, individual services, and geographic regions. Its key features include building cost charts based on various parameters, comparing current financial metrics with data from previous periods, and identifying the services or projects generating the highest expenses.

Furthermore, Google BigQuery Billing Export enables the export of detailed billing data to BigQuery, where SQL queries can be used to build flexible cost analytics, create reports, and perform custom cost forecasting (Vakhula et al., 2025).

### 5.5. Limitations of built-in services

Despite their significant functionality and ease of use, the built-in cost monitoring services provided by cloud vendors (AWS, Azure, GCP) have a number of fundamental limitations that reduce their effectiveness in complex multi-cloud scenarios.

Firstly, these services are tightly bound to a single provider's ecosystem, making it impossible to construct a consolidated view of costs in environments involving multiple cloud providers. The lack of cross-platform compatibility limits centralized financial monitoring capabilities and complicates the implementation of a FinOps approach in a multi-cloud context.

Secondly, there is an absence of a unified billing and reporting logic due to differences in cost presentation formats, data update frequencies, and levels of detail. This complicates comparative analysis and data integration from different sources. A third limitation is the basic level of automation. Although these tools can be integrated with services like AWS Lambda, Microsoft Power Automate, or Google Pub/Sub, they still

require additional development. Moreover, the standard functionality does not cover complex escalation scenarios, multi-level team interactions, or adaptation to internal organizational policies.

Finally, built-in services offer limited support for business context, as they mainly focus on the technical parameters of resources. This makes it difficult to map expenses to organizational structures (departments, teams, clients), business processes, or key performance indicators (KPIs), which are essential for strategic financial management (Vakhula and Opirskyy, 2024).

Built-in cost monitoring services are a first step toward controlling cloud budgets. They are well-suited for companies that use a single provider and want to establish basic control, alerting, and analytics rules. However, in the case of a multi-cloud strategy or the need for a centralized FinOps approach, these tools are generally insufficient. In such cases, independent multi-cloud solutions come into play, offering comprehensive integration, more flexible rules, automation, and financial analytics (Pavlenko et al., 2024).

### 5.6. Independent multi-cloud platforms for cost monitoring

These platforms implement the FinOps principles introduced in Section 2, enabling cost aggregation, threshold-based alerting, and automated escalation.

**Apptio Cloudability.** A leading FinOps platform for aggregating, analyzing, and optimizing costs across AWS, Azure, GCP, Kubernetes, (Samad et al., 2024) and on-premises infrastructure. It offers cost visualization by project, team, and service; spending forecasts; and optimization recommendations (e.g., switching to Reserved or Spot instances). Supports budget limit control, dashboards, automated reporting, and adaptive role-based reports for finance, technical, and DevOps teams (Deineka et al., 2024).

**Flexera One (formerly RightScale).**

An enterprise platform combining CMDB capabilities with cost monitoring and policy enforcement for cloud and on-premises resources. It enables cost breakdowns, benchmarking, and detection of inefficient usage with automated savings actions. Integration with ITSM tools (e.g., ServiceNow, Cherwell) supports automated incident handling and workflow approvals, making it well-suited for large, complex organizations (Brid, 2025).

**Spot by NetApp.**

A platform focused on real-time automated optimization of cloud resources to maximize cost efficiency while maintaining service performance. It automates switching between compute instance types, dynamically scales Kubernetes clusters (via Ocean), and provides cost-reduction recommendations — ideal for DevOps and SRE teams in highly dynamic environments (Harasymchuk et al., 2024).

**CloudHealth by VMware.**

A cloud financial management platform used by large enterprises and MSPs. It aggregates billing and operational data from major clouds, visualizes costs and KPIs, supports role-based access, integrates with compliance tools, and builds financial models to guide strategic deployment decisions across AWS, Azure, and GCP (Kumar 2025).

### 5.7. Comparison of multi-cloud platforms for cost monitoring

The conducted review and comparative analysis of independent multi-cloud platforms for cost monitoring demonstrates (Table 2) that the current market offers flexible and functionally rich solutions for implementing a FinOps strategy in IT organizations.

Each of the platforms reviewed has its own functional focus, which determines its intended purpose and role in the overall cloud resource financial management system. Specifically, Apptio Cloudability specializes in deep financial analysis, cost forecasting, and managing long-term financial commitments, making it particularly effective for strategic budget planning.

The Flexera One platform combines the capabilities of a CMDB system with FinOps analytics, providing not only cost control but also active resource management and the implementation of cost-saving policies at the organizational level. In turn, Spot by NetApp focuses on full automation of optimization processes, with a particular emphasis on Kubernetes environments, making it an attractive solution for DevOps teams in dynamic, scalable infrastructures.

The CloudHealth by VMware platform stands out with its advanced cost visualization system and business-oriented analytics, tailored to the needs of users at various levels — from technical specialists to financial directors and top management. It should be noted that all platforms, except for Spot, support cost forecasting and optimization recommendations. Meanwhile, only some solutions provide deep integration with ITSM systems and CMDB, which is a key factor for building a complete, managed FinOps ecosystem in a multi-cloud architecture (Venkatesh et al., 2025).

Thus, the choice of platform should be based on the strategic goals of the organization: whether it is cost control and reduction, integration with change processes, or the creation of a single source of truth for financial-technical decision-making.

### 5.8. Integrated solutions based on SIEM/CMDB

Integrated solutions based on SIEM or CMDB systems play a special role in building a systematic, process-driven, and controlled approach to managing costs in the cloud environment. These are not specialized FinOps-class platforms but allow for:

- Combining financial analytics with current IT processes.
- Integrating cost data into the incident, change, and configuration management structure.
- Running automated response scenarios in case of over-expenditure or policy violations.

The most common examples of such solutions are Splunk (as a SIEM/Observability platform) and Cherwell or ServiceNow (as CMDB/ITSM systems) (Pashikanti, 2025).

### 5.9. Splunk with cloud monitoring and billing add-ons

The Splunk platform is traditionally used as a SIEM tool for collecting, processing, and analyzing logs, events, and operational metrics in real time. However, thanks to specialized add-ons, Splunk can also be effectively used for monitoring financial metrics, particularly cloud costs.

Integration with major cloud providers — Amazon Web Services, Microsoft Azure, and Google Cloud Platform — is implemented through official Splunk add-ons (e.g., *Splunk Add-on for AWS Billing, Azure Monitor Add-on*), which allow automatic retrieval of data about costs, budgets, tags, instances, and billing alerts. The collected information can be visualized in customized analytics dashboards, where users can analyze expenses by projects, teams, or services, identify deviations from averages, and build integrated multi-cloud reporting (Samuel et al., 2024).

One of the key features of Splunk in cost monitoring is the ability to set custom thresholds for detecting overspending, followed by automatic alert generation or escalation scenarios. For this, a custom rule system is used, enabling reactions via webhook, email, or other integration mechanisms. Additionally, storing historical billing data opens opportunities for trend analysis and long-term cost forecasting. The main advantage of using Splunk as a cost monitoring platform is its high level of flexibility and scalability, as well as its ability to adapt to the specific needs of a corporate environment. However, using this tool requires initial knowledge of the Splunk Query Language (SPL), application configuration, and creating custom dashboards and response logic, which may require additional resources during the implementation phase (Thummala and Singh, 2025).

### 5.10. Cherwell and ServiceNow with billing data integration

The platforms Cherwell and ServiceNow belong to the ITSM/CMDB solutions class and are used for modeling IT infrastructure, managing changes, handling requests, incidents, and related service processes. In addition to traditional IT operations support functions, these systems are increasingly used for integrating financial information, particularly for monitoring cloud costs and budget control (Gupta and Singh, 2025).

Integration with cloud platforms (AWS, Azure, GCP) is implemented via APIs or using intermediary solutions (MuleSoft, Zapier, custom middleware components). This allows for the automatic synchronization of data about costs, budgets, responsible persons (e.g., CSA —Owner), daily and monthly resource consumption volumes, as well as tracking instances of exceeding established budget limits. This information is stored in configuration database objects and can be linked to specific projects, teams, or departments.

A key advantage of using Cherwell or ServiceNow in financial monitoring3 is the ability to automatically create incidents or requests when certain cost thresholds are reached. For example, when the use of a cloud project budget exceeds 90 %, the system generates a request to confirm whether to continue resource consumption. If no response is received within a specified period (e.g., 24 h), an incident is created and forwarded to the appropriate team (FinOps, DevSecOps, or IT department manager). These systems also support building approval workflows — processes for approval that can include requests for budget increases, access to additional resources, or consideration of non-standard expenses. Using response rules, users can configure who and when to send notifications, which escalations to initiate, and which events should be triggered automatically. Additionally, the platforms offer deep integration with communication and development tools such as email, Microsoft Teams, Slack, Jira, or DevOps pipelines (Yadav, 2025).

The advantage of such CMDB solutions is their high level of structure, role management, and the connection of financial information with real IT processes. They allow for embedding cost monitoring directly into the service and incident lifecycle, ensuring close collaboration between technical and financial functions within the organization. However, it should be noted that these platforms typically do not include deep cost analytics "out of the box" and require configuration of integrations, customization of response logic, and the development of custom reports or dashboards. Although Splunk, Cherwell, and ServiceNow are not classical FinOps platforms, they play an important role in providing an integrated approach to cost management in cloud environments. Thanks to their functionality, these systems allow not only for collecting and storing cost data but also for embedding financial event monitoring directly into operational IT processes.

Such solutions promote the development of a process-driven approach to cost control, where financial metrics are linked to the service lifecycle events — configuration changes, incidents, and service requests. This enables the creation of flexible response logic that considers organizational roles, Service Level Agreements (SLAs), and levels of responsibility within the IT structure (Malaraju, 2025).

These tools are particularly valuable for organizations that are already using Splunk or CMDB solutions, as they allow for the integration of cost management into the existing service and monitoring infrastructure, avoiding the need to deploy a separate FinOps platform. This approach enhances the effectiveness of cross-functional interaction between IT and finance departments, ensuring transparency, control, and accountability in the use of cloud resources.

### 5.11. Integrated solutions based on SIEM/CMDB for cost monitoring

In modern IT organizations, integrated platforms that combine monitoring, event management, configuration management, and service management capabilities are increasingly applied to cost observability in multi-cloud environments. These include SIEM systems such as Splunk and CMDB / ITSM platforms such as Cherwell and ServiceNow. While these solutions are not specialized FinOps tools, their ability to embed financial data into existing operational and security workflows enables flexible budget control and operational compliance.

As mapped in Section 2.2 (Table 1), these integrations directly support SOC 2 TSC by correlating cost anomalies with configuration and event data to detect, escalate, and mitigate financial risks. Rather than re-explaining SOC 2 concepts, this section focuses on how such integrations translate those criteria into measurable technical outcomes within multi-cloud governance (Pochu et al., 2024).

**SOC 2–Aligned use cases.**

The integration of SIEM and CMDB platforms into cost monitoring workflows enables operational scenarios that address specific SOC 2 criteria:

- **Security-driven anomaly detection** – correlation of cost anomalies with security event logs in Splunk to detect unauthorized resource provisioning or misuse (CC6.1, CC7.2).
- **Policy compliance enforcement** – automated creation of incidents in ServiceNow or Cherwell when budget thresholds linked to policy constraints are exceeded (CC7.3, PI1.1).
- **Availability protection** – CMDB prioritization of critical workloads to ensure they remain unaffected by budget exhaustion (A1.2).
- **Audit readiness and accountability** – combined SIEM/CMDB logging of cost-related alerts, escalations, and role-based approvals (CC7.4, CC6.6).

**Table 1**
Mapping of SOC 2 TSC to implemented controls and mitigated risks.

| SOC 2 Trust Services Criterion | Implemented Control in the System | Risk Mitigated |
|---|---|---|
| **CC6.1 Logical Access Security** | Role-Based Access Control (RBAC) in Splunk and Cherwell; restricted modification of budget parameters | **Unauthorized resource provisioning** — prevention of changes by unauthorized personnel |
| **CC6.6 Separation of Duties** | Segregation of responsibilities between FinOps, DevSecOps, and SecOps for change approvals and incident resolution | **Unauthorized resource provisioning** — mitigation of risks from compromised accounts or bypassed approval workflows |
| **CC7.2 Detection and Monitoring of System Components** | Automated detection of untagged or unapproved workloads via cost anomaly patterns | **Untracked or shadow workloads** — identification of uncontrolled resources and associated expenditures |
| **A1.2 Availability Commitments** | Escalation workflows when budget thresholds risk exhaustion; prioritization of critical workloads | **Budget exhaustion** — prevention of service interruptions and SLA breaches |
| **CC7.3 Incident Response** | Threshold-based alerting in Splunk, automated escalation via Cherwell/ ServiceNow | **Ineffective alert configuration** — ensures timely detection and response to anomalous cost activity |
| **PI1.1 Processing Integrity** | Regular ingestion and validation of cost data from AWS, Azure, GCP APIs | **Ineffective alert configuration** — prevention of false negatives due to incomplete or inaccurate data |
| **CC7.4 Logging and Monitoring of Activities** | Centralized logging of budget breaches, escalations, and resolutions in Splunk | **Misuse of resources** — creation of an audit trail to detect and investigate unauthorized use |
| **CC9.2 Confidentiality of Information** | Controlled access to financial reports and cost allocation data | **Misuse of resources** — limiting exposure of sensitive cost data that could enable exploitation |

**Table 2**
Comparison of cost monitoring systems.

| Platform | Multi-Cloud Support | Automation | Integration with ITSM/ CMDB | Cost Forecasting | Security Features |
|---|---|---|---|---|---|
| **Apptio Cloudability** | AWS, Azure, GCP, Kubernetes, On-Prem | Yes (recommendations, rights-sizing, commitments) | Indirectly | Yes | Role-based access control, encrypted data storage, supports SOC 2 alignment |
| **Flexera One** | AWS, Azure, GCP, On-Prem | Yes (overspending detection, policies) | Yes (e.g. ServiceNow) | Yes | Strong policy management, CMDB integration, access governance |
| **Spot by NetApp** | AWS, Azure, GCP, Kubernetes | Yes (auto-optimization Spot/ Reserved, Kubernetes) | No | No | Automated scaling with cost-based policies, identity-aware usage control |
| **CloudHealth by VMware** | AWS, Azure, GCP | Partially (through integrations and scenarios) | Yes | Yes | Supports compliance (SOC 2, HIPAA), secure API access, audit logs |

These use cases demonstrate that SIEM and CMDB platforms can serve as integral components of a SOC 2–aligned control environment, ensuring that cost anomalies are governed with the same rigor as other security incidents. (Guerbaoui et al., 2025)

## 6. SOC 2–aligned risk coverage through SIEM/CMDB integration

The risk scenarios identified in Section 2.2 are addressed through the integration of SIEM and CMDB functionalities into cost monitoring processes. By correlating financial and operational telemetry, these systems provide preventive and detective controls that directly reduce the likelihood and impact of identified threats. Table 3 summarizes this alignment, showing each SOC 2 criterion with its implementation example, before/after conditions, and quantitative improvement (Patwardhan and Karim, 2025).

This SOC 2–aligned mapping confirms that financial observability, when embedded into SIEM and CMDB platforms, extends beyond cost governance into the domain of operational security. By explicitly linking risks to TSC and quantifying improvements, the table demonstrates how integration supports both preventive and detective controls. Each scenario not only strengthens SOC 2 compliance but also enhances the organization's ability to detect, contain, and remediate incidents with financial and security implications. This structured approach ensures that cost-related risks are governed within the same control environment as other critical security processes, thereby increasing both audit readiness and operational resilience (Shah et al., 2025).

These solutions are particularly effective for organizations that already have an established service management system or use Splunk as a SIEM/Observability platform. In such cases, integrating financial monitoring into existing IT processes is a logical and cost-effective step that ensures high transparency, accountability, and control over the use of cloud resources (Singh, 2025).

## 7. Description of the proposed solution for cost monitoring in a multi-cloud environment

The proposed solution is an integrated cost control system for multi-cloud infrastructure, combining capabilities for monitoring, analysis, alerting, and automated response to budget overruns. The architecture is based on the Splunk platform (for collecting, aggregating, and analyzing billing data) and Cherwell (as the ITSM/CMDB system for storing data on budgets, responsible parties, incident management, and automation of response processes).

The system is designed to identify overspending scenarios in near real time and escalate them through a multi-level logic of alerts and responsibilities. As shown in Eq. (1), the system continuously evaluates the ratio of actual costs $C(t)$ to the approved budget $B$, triggering escalation events $T_k$whenever the threshold $\theta_k$ is exceeded (Varadaraj, 2025).

Eq. (2) defines the behavior when no timely response is received from the responsible stakeholder. If budget consumption exceeds 110 % and no response occurs within the time window $\tau$, the incident is escalated to the DevSecOps team. From Eq. (3), it follows that if the absolute deviation $\delta(t)$ exceeds the predefined overrun margin $\gamma$, an incident is raised to the SecOps team regardless of prior escalation or acknowledgment.

The notification logic is formalized in Eq. (4) as a piecewise function $N(t)$, ensuring that the right stakeholders — including the budget owner, project manager, and CSA — are informed at appropriate escalation stages.

A key feature of the solution is its adaptability to any cloud environment (AWS, Azure, Google Cloud) and its ability to operate with multiple providers simultaneously, providing a single point of analytics and control.

**Table 3**
Enhanced mapping of SOC 2 control improvements through SIEM/CMDB integration.

| SOC 2 Criterion | Before Implementation | After Implementation | Improvement/ Coverage Impact |
|---|---|---|---|
| **CC7.3 Incident Response** | New resources could be provisioned without detection due to ineffective alerting | Splunk alert triggers a ServiceNow incident when a new resource appears without an approved CMDB change record; escalation automatically assigned to the responsible DevSecOps team. | Reduces manual oversight by ≈50 %; accelerates escalation of unapproved deployments and improves incident traceability. |
| **CC6.6 Change Management** | Shadow workloads remain untracked, leading to uncontrolled costs and non-compliance | CMDB reconciliation with billing data automatically identifies workloads not linked to registered assets; unresolved mismatches generate compliance tickets. | Ensures 100 % asset visibility; reduces configuration drift and compliance gaps by ≈40 %. |
| **A1.2 Availability** | Budget exhaustion could cause disruption of critical workloads | CMDB asset prioritization triggers Splunk alerts to scale down non-critical instances when forecast breach > 90 %. | Maintains SLA compliance; minimizes service disruption risk by ≈35 %. |
| **CC7.2 Monitoring** | Misconfigured alerts delay anomaly detection | SIEM correlation rules flag when cost alert channels fail or misfire; automatic verification every 6 h. | Shortens response time by ≈45 %; improves detection reliability and SOC coverage. |
| **CC7.4 Audit Logging** | Misuse of resources may remain invisible due to lack of traceability | Combined Splunk and CMDB logs create a complete audit trail of anomalies, notifications, and escalations; data retention ≥ 1 year. | Enables forensic investigation; supports SOC 2 Type II audit readiness with full evidence chain. |

*7.1. Advantages of the proposed solution. The developed cost monitoring system architecture has a number of advantages that ensure its effectiveness in complex multi-cloud environments*

First, the solution is fully multi-cloud, meaning it provides unified monitoring and analytics for different cloud providers (AWS, Azure, GCP), allowing it to avoid data fragmentation and create a single information environment for decision-making.

Second, strong emphasis is placed on process automation, including data collection, threshold calculation, notifications, incident creation, and escalation. This minimizes manual intervention, reduces response time, and increases system reliability (Mehta, 2021).

The third advantage is the clearly structured escalation model, which ensures cost control at each stage of growth, considering the level of criticality and the responsibilities of the involved participants.

The fourth advantage lies in integration with business roles: the system accounts not only for technical parameters but also organizational authority, enabling financial, operational, and technical teams to jointly participate in decision-making regarding cloud resource usage.

Finally, an important functionality is cost forecasting, which is implemented through integration with Splunk. This enables the generation of analytical reports, trend identification, and assessment of future resource needs — forming the basis for strategic budget planning (Smith and Ok, 2025).

**Mathematical model for threshold calculation and deviation analysis in multi-cloud environments.**

To effectively manage cloud expenditures and support timely incident escalation, the following mathematical model is proposed. It provides a structured method for calculating budget thresholds, detecting deviations, and initiating response workflows based on financial anomalies.

*Notation:*

- $B$:Approved budget for a given project over a defined time period (e.g., monthly), measured in monetary units.
- $C(t)$:Actual cloud expenditure at time $t$.
- $\delta(t) = C(t) - B$: Absolute deviation from the budget at time $t$.
- $\theta_k \in (0,1]$: Budget threshold for escalation level $k$, where:
- $\theta_1 = 0.7$ : Yellow level — early warning;
- $\theta_2 = 0.9$ : Orange level — elevated risk;
- $\theta_3 = 1.0$ : Red level — full budget consumption or overspend.
- $\gamma$ : Absolute overage threshold for triggering a critical incident (e.g., USD 2000).
- $T_k$: Escalation event triggered at threshold $\theta_k$
- $\tau$: Maximum allowable time (in hours or days) without response from a responsible party (e.g., CSA).

**Threshold-based escalation events.** For each escalation level $k \in \{1,2,3\}$, the following condition is evaluated:

$$\frac{C(t)}{B} \geq \theta_k \tag{1}$$

If this inequality holds, an event $T_k$ is triggered, initiating the corresponding notification and response workflow.

*7.1.1. Escalation in the absence of timely response. If the system detects that*

$$\frac{C(t)}{B} \geq 1.1 \text{ and } \Delta t \geq \tau \tag{2}$$

where $\Delta t$ is the duration without acknowledgment or corrective action from the responsible party, an incident is escalated to the **DevSecOps** team.

*7.1.2. Critical overrun detection. If the absolute deviation exceeds the predefined critical threshold*

$$\delta(t) = C(t) - B \geq \gamma \tag{3}$$

an incident is immediately raised to the **SecOps** team, regardless of prior response status.

*7.1.3. Notification function. A piecewise notification function can be formalized as*

$$N(t) = \begin{cases} \text{send_notice}(T_1, \text{BudgetOwner}), & \text{if } \theta_1 \leq \frac{C(t)}{B} < \theta_2 \\ \text{send_notice}(T_2, \{\text{BudgetOwner, ProjectManager}\}), & \text{if } \theta_2 \leq \frac{C(t)}{B} < \theta_3 \\ \text{send_notice}(T_3, \{\text{BudgetOwner, ProjectManager, CSA}\}), & \text{if } \frac{C(t)}{B} \geq \theta_3 \end{cases} \tag{4}$$

This function ensures that the right stakeholders are informed based on the severity of budget consumption.

*7.1.4. Cumulative deviation metric. To monitor the aggregate impact of spending over time, an integral deviation metric can be used*

$$D = \int_0^T \left| C(t) - B \right| dt \tag{5}$$

Alternatively, for systems with discrete time intervals (e.g., daily billing), a summation form is appropriate:

$$D = \sum_{i=1}^{n} \left| C_i - B \right| \tag{6}$$

This provides insight into long-term deviations from the planned budget.

Forecasting model. Forecasting cloud expenditures is essential for proactive budget management and timely anomaly detection. In the context of cost observability as a security control, predictive models enable organizations to anticipate threshold violations in advance, allowing for early mitigation actions and reducing the likelihood of uncontrolled overspending.

Linear cost forecasting. Assuming that cloud expenditure increases approximately linearly over time (a common scenario in stable workloads), a simple linear regression model can be used to forecast future costs:

$$\widehat{C}(t) = \alpha t + \beta \tag{7}$$

Where:

- $\widehat{C}(t)$ is the predicted cumulative cost at time $t$,
- $\alpha$ is the estimated daily (or hourly) cost growth rate,

- $\beta$ is the intercept, representing initial accumulated cost (often 0 at the beginning of the billing cycle),
- $t$ is the time variable (in days or hours since the start of the billing period).

This model can be fit using historical data over the current billing cycle (e.g., first 7–10 days of the month) using ordinary least squares regression.

**Threshold time estimation.** To estimate when a specific cost threshold $\theta_k \cdot B$ will be breached, we solve the equation:

$$\widehat{C}(t_k^*) = \theta_k \cdot B \tag{8}$$

Solving for $t_k^*$:

$$t_k^* = \frac{\theta_k \cdot B - \beta}{\alpha} \tag{9}$$

This expression gives the estimated time (in days or hours) at which the cost will reach the threshold level $\theta_k$ (e.g., 70 %, 90 %, or 100 % of the allocated budget $B$).

**Forecast accuracy and error margin.** To improve reliability, the model can incorporate a confidence interval using standard linear regression error analysis. For example, the upper bound forecast can be defined as:

$$\widehat{C}_{\text{upper}}(t) = \widehat{C}(t) + z \cdot \sigma \tag{10}$$

Where:

- σ is the standard error of the forecast,
- $z$ is the z-score corresponding to the desired confidence level (e.g., 1.96 for 95 %).

By comparing both lower and upper bounds with defined thresholds, the system can trigger pre-emptive warnings even before an actual breach occurs.

#### 7.1.5. Extensibility to other cloud service providers

Although the proposed system was evaluated using the three largest public cloud providers (AWS, Azure, and GCP), the architecture is designed to be extensible to additional Cloud Service Providers (CSPs). The core requirement is the availability of standardized billing and usage data that can be programmatically retrieved and normalized. Specifically, integration with new CSPs requires:

**Accessible billing APIs** – the CSP must expose programmatic access to cost and consumption data in a structured format (e.g., JSON, CSV, or REST-based endpoints).

**Granularity of resource metadata** – the billing data should include resource identifiers, tags, or project associations, enabling alignment with CMDB assets and cost allocation policies.

**Event streaming or export capabilities** – to ensure near real-time anomaly detection, the CSP should support scheduled exports or event-driven integration into SIEM and ITSM platforms.

**Security and compliance transparency** – the provider must deliver audit-ready cost and usage records that can be mapped to SOC 2 TSC, ensuring continuity of compliance coverage.

Given these prerequisites, the solution can be extended to emerging providers such as Oracle Cloud Infrastructure (OCI), IBM Cloud, or specialized regional CSPs, provided they supply programmatic billing access and metadata granularity comparable to AWS, Azure, and GCP.

Alternative forecasting methods for improved accuracy

The baseline forecasting approach applied in this study relies on linear regression, as detailed in Section 4.1.7. While this model provides interpretability and computational simplicity, it assumes a constant rate of cost increase over time. Such an assumption is often unrealistic in multi-cloud environments, where cost behavior may reflect nonlinear trends, seasonal patterns, or dependencies on multiple operational factors.

To address these limitations, two alternative methods were considered: Random Forest Regression and Kernel Regression. These models provide flexibility in capturing nonlinear cost dynamics and are increasingly adopted for operational forecasting in complex infrastructures.

Random Forest Regression

Random Forest Regression is an ensemble learning method that constructs multiple decision trees and averages their outputs to produce a final prediction. In the context of cloud cost forecasting, input features can include:

- Time index (day, week, month)
- Number of active virtual machines (VMs)
- Distribution of resource types (On-Demand, Reserved, Spot)
- User activity metrics
- Seasonal indicators (month, quarter)

Advantages:

- Captures complex non-linear patterns and feature interactions.
- Robust to noise and outliers.
- Provides variable importance measures, allowing identification of cost-driving factors.

SOC 2 Relevance:

The multi-factor analysis capability supports improved anomaly detection for CC7.2 (Detection and Monitoring) and PI1.1 (Processing Integrity) by correlating financial anomalies with operational metrics.

#### 7.1.6. Kernel Regression (Nadaraya–Watson estimator)

Kernel Regression is a non-parametric approach that estimates the value at a given point as a weighted average of surrounding observations, with weights determined by a kernel function K(·):

$$\widehat{C}(t) = \frac{\sum_{i=1}^{n} K\left(\frac{t-t_i}{h}\right) \cdot C(t_i)}{\sum_{i=1}^{n} K\left(\frac{t-t_i}{h}\right)} \tag{11}$$

where h is the bandwidth parameter controlling the smoothness of the curve.

Advantages:

- Adapts to local variations in cost data.
- Effective for datasets with seasonality or burst patterns.
- Requires no assumption of functional form.

SOC 2 Relevance:

Enhanced forecast precision aids in preventing budget exhaustion, aligning with A1.2 (Availability Commitments) and reducing the risk of service disruption.

**Comparison of forecasting methods for improved accuracy**

The comparison is based on well-established error metrics commonly used in predictive analytics:

·**Root Mean Squared Error (RMSE):** Measures the square root of the average squared differences between observed and predicted values. Larger errors are penalized more heavily, making RMSE sensitive to significant deviations.

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} (y_i - \widehat{y}_i)^2}{n}} \tag{12}$$

·**Mean Absolute Error (MAE):** Represents the average of absolute differences between observed and predicted values. It provides a straightforward interpretation of the typical prediction error.

$$MAE = \frac{\sum_{i=1}^{n} |y_i - \hat{y}_i|}{n} \tag{13}$$

·**Coefficient of Determination** ($R^2$): Indicates the proportion of variance in the observed data explained by the model. A value closer to 1 implies higher predictive accuracy.

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \tag{14}$$

To provide a stable benchmark, the models were applied to the average monthly costs across all twelve months of 2024.

- Observed Cost (average monthly costs): ≈ 97,560 $
- Forecast Linear (average): ≈ 118,982 $
- Forecast Random Forest (average): ≈ 117,792 $
- Forecast Kernel (average): ≈ 117,197 $

The average monthly dataset was then used to calculate error metrics across the 2024 period (Tables 3, 4).

The evaluation demonstrates that advanced forecasting models such as Random Forest and Kernel Regression provide measurable accuracy gains over the linear baseline, particularly in capturing non-linear behaviors and seasonal cost fluctuations. Random Forest achieved the highest accuracy, while Kernel Regression showed robustness under conditions of burst-type workloads.

However, linear regression remains a valuable component of the evaluation framework. Its simplicity, transparency, and low computational overhead make it highly suitable as a baseline model and as a tool for rapid cost projections in environments with limited historical data. Furthermore, the direct interpretability of its coefficients facilitates communication with financial stakeholders and auditors, aligning with SOC 2 requirements for clarity and accountability Fig. 1.

Thus, the results suggest a tiered forecasting approach:

- Linear regression for fast, interpretable baseline estimates.
- Random Forest or Kernel Regression for improved accuracy and resilience in complex, dynamic multi-cloud environments.

This layered methodology balances interpretability, computational efficiency, and predictive accuracy, reinforcing both financial observability and SOC 2–aligned risk management.

The principles outlined above are realized through the architectural components described in Section 4.2

*7.1.7. Description of the functional components of the architectural solution. One of the key advantages of the proposed system is its modular structure, which allows the integration of data sources, the analytical core, the event management center, and the automated alerting system into a unified platform for cost monitoring in a multi-cloud environment*

At the data source level, the primary providers of information about cloud expenditures are Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP). Data on resource consumption and their costs are imported via official APIs in JSON format. The update frequency is set to three times a day, which enables near real-time monitoring of financial indicators (Madupati, 2025).

The central analytical component of the system is the Splunk platform, which performs the functions of aggregating and processing cost data. Specialized add-ons are used for this purpose, including the GCP Splunk Add-on, AWS Splunk Add-on and Azure Splunk Add-on, which allow the retrieval and processing of billing information from the respective cloud platforms. Within Splunk, algorithms are implemented for aggregating expenses by provider, project, and budget item, as well as for calculating the proportion of actual consumption within the established budget. The system also supports a multi-level escalation model based on predefined threshold values: 70 % (conditionally "yellow" level), 90 % ("orange"), and 100 % or budget overrun ("red").

Eqs. (1)–(4) collectively describe the logic used to determine escalation triggers and stakeholder notifications

These computational models establish the predictive basis for the escalation mechanism that translates financial anomalies into actionable workflows within the cost governance process.

Upon reaching each threshold, service notes may be created and notification scenarios for responsible parties initiated.

**Escalation as a multi-stage cost control mechanism**

Escalation in this system is designed not only as a technical trigger but as a structured multi-stage governance mechanism. It ensures that cost overruns are handled consistently through defined levels of responsibility and automated workflows.

- **Informational level (70 % of budget):** notifications are sent to budget owners, allowing them to review consumption trends.
- **Operational level (90 %):** the system automatically generates service requests in Cherwell or ServiceNow, requiring validation of workloads or budget adjustment.
- **Critical level (100 %+):** high-priority incidents are escalated to DevSecOps and financial controllers, with safeguards such as suspending shadow workloads or reallocating resources.

This layered structure ensures early awareness, operational intervention, and executive accountability. By aligning escalation with SOC 2 TSC (CC6.1, CC7.2, PI1.1), the system not only prevents uncontrolled financial growth but also enforces systematic accountability, making financial anomalies an integral part of security and compliance monitoring.

Role in growth phases

This multi-level escalation model enables organizations to maintain financial discipline during growth stages, when the number of accounts and workloads increases rapidly. Escalation ensures that:

- Initial overages are treated as early warnings, allowing teams to react without disruption.
- Repeated or significant deviations are formalized into controlled processes, preventing uncontrolled cost accumulation.
- Critical overruns are handled with executive visibility and automated safeguards, maintaining both budgetary control and service availability.

By embedding escalation into cost monitoring, financial anomalies are not only detected but systematically governed, thereby aligning cost control with SOC 2 requirements for accountability, consistency, and incident management.

**Escalation and automated response workflow**

Information about budget constraints, responsible individuals, and expense statuses is stored in the Cherwell system, which serves as a metadata repository and response center for overspending events. Cherwell records the limits of approved budgets, project identifiers, data on responsible individuals (CSA), and the current financial state of each monitored object. When spending reaches 90 %, the system automatically generates a service request. If actual budget overrun exceeds 110 %

**Table 4**
Example of comparison of forecasting methods.

| Model | RMSE ↓ | MAE ↓ | $R^2$ ↑ | Note |
|---|---|---|---|---|
| Linear Regression | 11,532 | 9847 | 0.872 | Interpretable but limited for nonlinear shifts |
| Random Forest Regression | 9885 | 8126 | 0.913 | Best accuracy with multi-factor dependencies |
| Kernel Regression | 10,143 | 8452 | 0.901 | Robust against seasonal fluctuations |

(↓ lower is better, ↑ higher is better).

| Project | Budget owner | Cloud type | Curent budget | Cloud threshold | Project manager | Cloud Owner | Responsible DevOps | Last Query update |
|---|---|---|---|---|---|---|---|---|
| Project 1 | Tony Stark | AWS | 900$ | 1000$ | Black Widow | Peter Parker | Felicia Hardy | 15h |
| Project 2 | Tony Stark | Azure | 100$ | 200$ | Captain America | Doctor Strange | Miles Marales | 23h |
| Project 3 | Tony Stark | GCP | 1000$ | 900$ | Adam Warlock | Kamala Khan | Wade Winston | 2h |

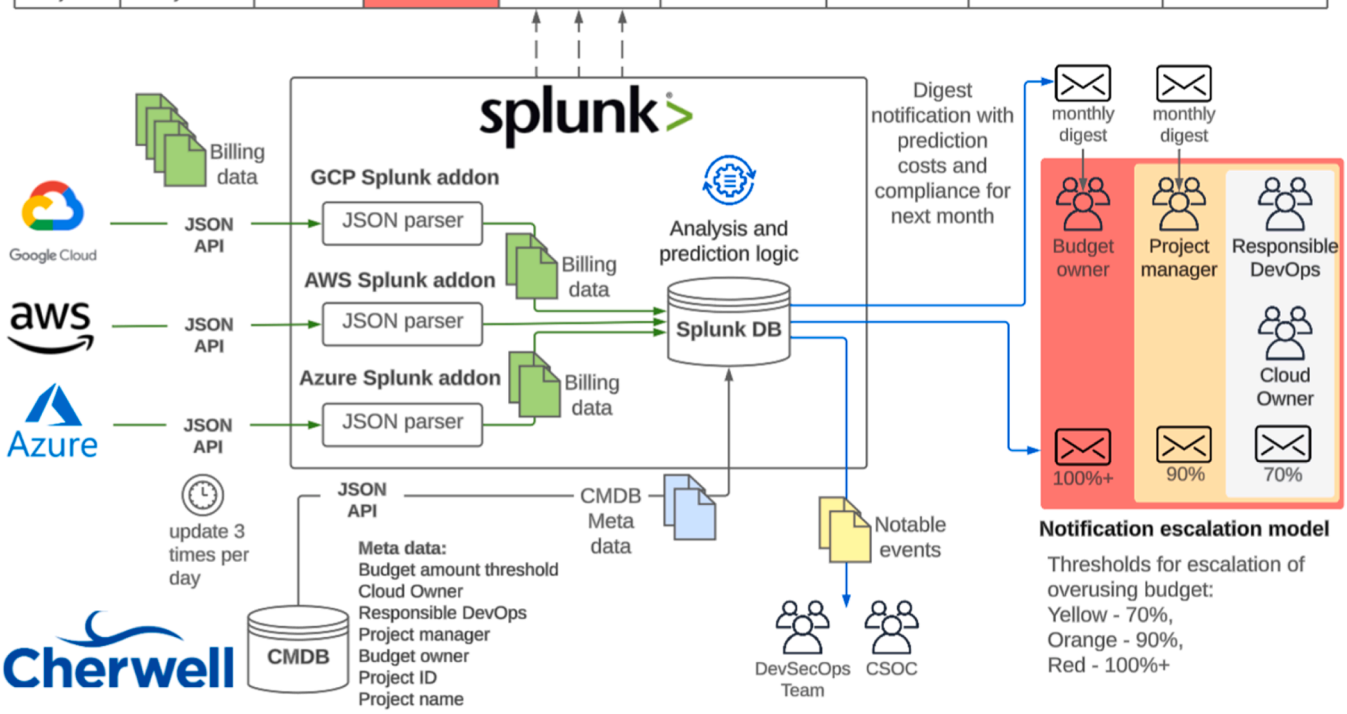


**Fig. 1.** Solution architecture with functional components.

and no response is received from the responsible user within one business day, an incident is created and forwarded to the DevSecOps team for processing. If the budget is exceeded by $2000 above the set limit, an incident is initiated and directed to the information security system (SecOps) (Perumal and Ahire, 2025).

A functionally important part of the architecture is the automated notification mechanism, which ensures timely communication with the relevant roles. Email alerts are sent when the thresholds of 70 %, 90 %, and 100 % are reached. Notifications are generated in a differentiated manner according to levels of responsibility: at the individual account level — for the budget owner, and at the organizational level — for the CSA or department head. Additionally, the system generates a monthly digest that includes a summary of current expenses, trend analytics, and forecast indicators based on models built in Splunk (Shelke and Frantti, 2025).

To proactively prevent budget overruns, the system utilizes a linear cost forecasting model, described in Eq. (7). This model estimates the expected cumulative cost at a given time based on the current rate of expenditure growth (denoted by coefficient α (alpha) and the initial accumulated cost at the beginning of the billing period (represented by parameter $\beta$.

Next, the system calculates the estimated time $t_k^*$ at which a predefined budget threshold (e.g., 70 %, 90 %, or 100 %) will be exceeded. This is done using Eq. (9), which solves for the point in time when forecasted costs are projected to cross the threshold $\theta_k$

To enhance reliability and account for uncertainty, the system also computes an upper bound for spending, incorporating statistical error margins as defined in Eq. (10). This includes the standard deviation of the forecast and a confidence interval (expressed via a z-score), allowing the system to detect risk patterns even before actual spending reaches critical limits.

In summary, this logic enables the system not only to detect cost anomalies but to anticipate them in advance, thereby improving budget planning and reducing the likelihood of financial incidents.

The system's operational logic is based on principles of maximum automation to minimize manual intervention. Regular data updates, automatic creation of notes, incidents, and requests, as well as built-in escalation rules, ensure the system's resilience and predictability in responding to changes in financial conditions.

To support proactive budget management, the described mechanisms are complemented by forecasting and anomaly detection models, which are further validated through empirical evaluation in Section 5.

#### 7.1.8. *Query examples for implementation*

To illustrate the operationalization of escalation and monitoring logic, the architecture can be extended with Splunk and SQL queries that transform abstract thresholds into actionable controls. These queries serve as templates for real-world deployment, supporting anomaly detection, incident creation, and audit logging in alignment with SOC 2 requirements.

**Splunk query for budget threshold detection**: index=cloud_costs
| eval budget_consumption_ratio = cost_amount / approved_budget
| where budget_consumption_ratio ≥ 0.7
| eval escalation_level = case(
budget_consumption_ratio ≥ 1.0, 3,
budget_consumption_ratio ≥ 0.9, 2,
budget_consumption_ratio ≥ 0.7, 1
)
| stats sum(cost_amount) as total_cost by provider_id, escalation_level

This query calculates the ratio of consumed to approved budget, assigns an escalation level (1 – informational, 2 – operational, 3 – critical), and aggregates the costs by provider. The output can be directly integrated with ServiceNow or Cherwell to trigger role-based incidents.

**SQL query for shadow workload detection**:
SELECT b.project_id, b.resource_id, b.cost_amount
FROM billing_data b

```
LEFT JOIN cmdb_assets c
ON b.resource_id = c.resource_id
WHERE c.resource_id IS NULL
AND b.billing_date BETWEEN '2024–01–01' AND '2024–01–31'
AND b.cost_amount > 100;
```

This SQL query compares billing data with CMDB assets to identify workloads that are generating costs but are not registered in the official asset database. Such anomalies represent potential shadow IT and uncontrolled expenditures, which can be escalated as compliance incidents.

In practice, these queries would be scheduled at regular intervals (e.g., every 6 h for billing checks, daily for CMDB reconciliation). While frequent execution increases operational visibility, it also introduces runtime costs in terms of API calls and data indexing. Organizations must balance detection granularity with resource efficiency.

#### 7.1.9. Example of a practical overspending response scenario

To demonstrate the operating logic of the proposed system, let us consider a typical situation where expenses for a specific project hosted on Google Cloud Platform (GCP) exceed the approved budget. Suppose the resource usage limit is set at $900, but actual consumption has reached 111 %, amounting to $1000.

In such a case, the system logs the threshold breach event in the Splunk analytics module. According to the built-in escalation logic, this corresponds to the "red level" of criticality, indicating a budget overrun. Splunk immediately initiates the sending of a notification to the responsible person — the owner of the CSA.

If no response is received or corrective actions are not taken within 24 h of the notification, the system automatically creates an incident in the Cherwell platform, which is then submitted for review by the DevSecOps team. If the overspending continues to grow and exceeds the approved budget by another $2000, the system initiates the creation of a new incident, this time at the SecOps level, which is responsible for security policies and strategic risk management.

This scenario demonstrates a closed-loop cycle of automated cost control, combining technological, organizational, and financial components of the response process.

It is also worth noting that the developed solution not only enables FinOps control but also facilitates the implementation of certain control procedures as part of SOC 2 audit preparation, including action logging, compliance with established SLAs, response to budget policy violations, and event history retention for future audit review.

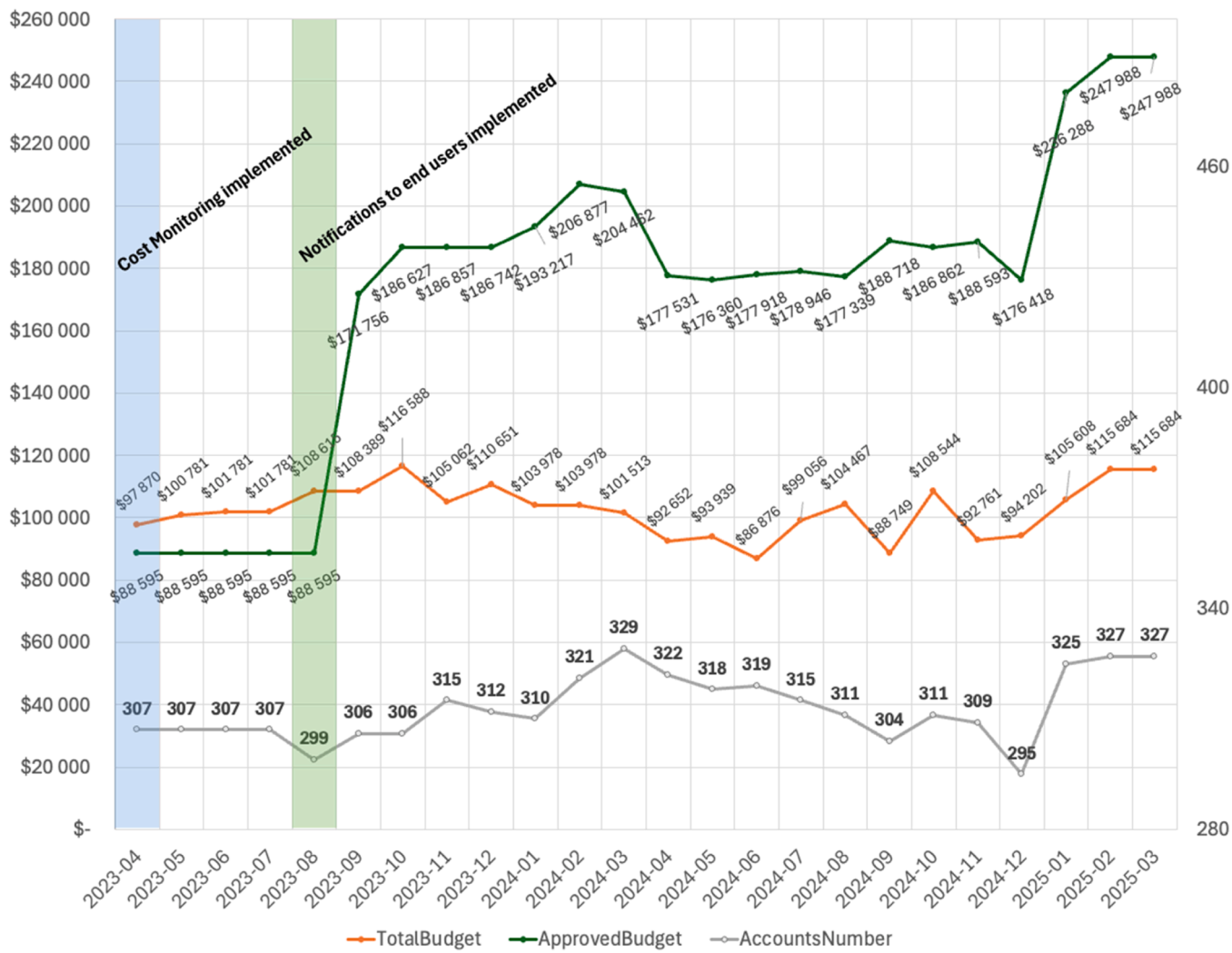


**Fig. 2.** Dynamics of multi-cloud environment costs from 2023 to 2025. The spike in July–August 2023 reflects the discovery of previously untracked resources and billing reconciliation following system implementation (see Section 5.2).

## 8. Results of implementing the cost monitoring system in a multi-cloud environment. To assess the effectiveness of the implemented solution, an analysis of cost dynamics in the multi-cloud environment was conducted for the period from April 2023 to March 2025

The evaluation follows the forecasting and escalation logic formalized in Eqs. (1)–(10), where cost dynamics C(t) are analyzed against approved budgets B and predefined thresholds $\theta_1$–$\theta_3$

A visualization of the results is presented in the graph «Cost monitoring for Multi-Cloud infrastructure» (Fig. 2).

General description of the graph. The graph illustrates changes in three main indicators:

- Total Budget (green line, left Y-axis) — total expenditures on cloud infrastructure.
- Approved Budget (orange line, left Y-axis) — the approved budget for the corresponding period.
- Accounts Number (gray line, right Y-axis) — the number of active accounts using cloud resources.

The graph also displays two vertical-colored zones indicating key implementation milestones:

- Blue zone (2023–04 to 2023–07): the period of cost monitoring system implementation.
- Green zone (2023–08): the beginning of automatic notifications to end users in cases of overspending.

### 8.1. *Cost and account number dynamics. Before the implementation of the system (first half of 2023), the total budget remained relatively stable, fluctuating within the range of $90,000–$100,000. At the same time, the approved budget had a clearly defined fixed limit (~$85,000–$89,000). The number of accounts also remained stable at around 307*

After the launch of monitoring in 2023–07, a sharp spike in the Total Budget is observed — up to $147,056. Then, in 2023–08, following the introduction of automatic notifications, spending increases even further, surpassing $185,621, while the Approved Budget rose only slightly (~ $109,839). This may indicate accumulated debt uncovered by the monitoring system or the emergence of previously hidden (shadow) expenses.

Throughout 2023–2024, the Total Budget gradually stabilizes within the range of $180,000–$200,000, with minor fluctuations. The Approved Budget shows a steady, moderate increase in line with the pace of infrastructure scaling. The total number of accounts also varies within the range of 304–329, with a decrease in August 2024 (to 295 accounts), followed by an increase to 327 by the beginning of 2025.

In the first quarter of 2025, another spike in expenditures is observed — the Total Budget reaches $247,988, which may be related to system scaling or increased cloud provider rates. At the same time, the Approved Budget also rises to $155,684, indicating an adaptation of financial planning to changes in consumption structure.

This anomaly, visible as a step increase in the green zone on Fig. 2, was traced to the identification of shadow resources and delayed billing adjustments rather than genuine usage growth.

#### 8.1.1. *Evaluation of monitoring effectiveness*

The implementation of cost monitoring and user notifications demonstrated a measurable effect on budget stability, forecast accuracy,

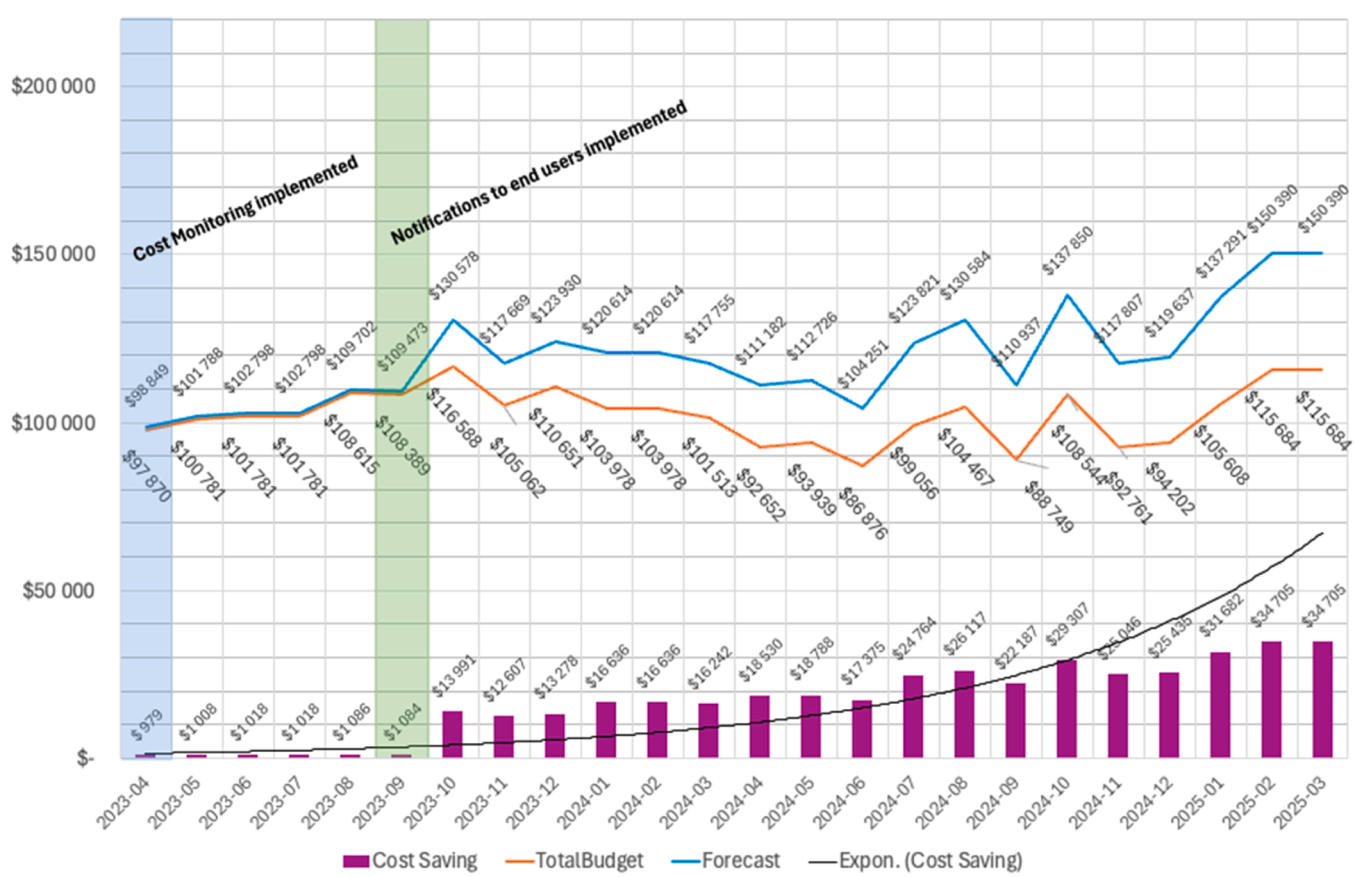


**Fig. 3.** Dynamics of cloud cost monitoring, forecasting, and savings (2023–2025). Savings calculated as difference between baseline consumption trajectory and actual spending following interventions (methodology detailed in Section 5.3.1).

and financial efficiency (Fig. 3).

Two critical interventions shaped the evaluation period. These improvements correspond to the forecast model described in Section 4.2, which links cost growth rate α and budget thresholds $\theta_1$–$\theta_3$ to proactive alert generation.:

·Cost Monitoring Implemented (September 2023):

After monitoring was introduced (blue zone), cost visibility improved significantly. Forecast values (blue line) started to align more closely with the actual total budget (orange line), reducing discrepancies and establishing a reliable baseline for financial analysis.

·End-User Notifications Implemented (November 2023):

Following the rollout of notifications (green zone), user behavior adapted, leading to more consistent consumption patterns. This reduced anomalies, stabilized budget dynamics, and reinforced accountability at the operational level.

### 8.2. Cost savings from proactive monitoring

The introduction of systematic monitoring and user notifications resulted not only in enhanced transparency but also in measurable financial outcomes. In the initial phase, corresponding to mid-2023, cost savings were marginal and did not exceed $2000 per month. However, as the system matured and user behavior adapted, a steady increase in efficiency was observed throughout 2024, with monthly savings frequently surpassing $15,000–$20,000. By early 2025, the savings exceeded $34,000 per month. The exponential trend line depicted in Fig. 3 underscores the cumulative efficiency gains attributable to proactive monitoring and corrective actions initiated through notifications.

#### 8.2.1. Savings calculation methodology

The cost savings presented in Fig. 3 represent the financial impact of proactive monitoring and user behavioral changes following system implementation.

Savings are computed as:

$$\text{Savings}(t) = C_{\text{baseline}}(t) - C_{\text{actual}}(t)$$

where $C_{\text{baseline}}(t)$is the projected cost assuming pre-implementation consumption patterns, and $C_{\text{actual}}(t)$is observed spending with active monitoring.

The baseline rate $\alpha_{\text{baseline}}$was derived from April–June 2023 data (average 2.3 % monthly growth):

$$C_{\text{baseline}}(t) = C_0 \times (1.023)^t$$

with $C_0 = \$98,450$(average monthly cost, Q2 2023).

**Sources of savings:**

1. Elimination of Orphaned Resources (42 %) — idle EC2 instances, unattached EBS volumes, excess snapshots.
2. Right-Sizing (31 %) — instance optimization based on 90 % alerts.
3. Reserved Instance Optimization (18 %) — conversion from On-Demand.
4. Early Termination of Test Environments (9 %) — automatic shutdown at 70 % threshold.

Validation was conducted using Cherwell incident logs, AWS/Azure/GCP billing reports, and financial reconciliation records.

The cumulative savings (July 2023 – March 2025) totaled $389 400 — a 22.7 % reduction compared to the baseline projection.

**Attribution of savings.**

To ensure that savings resulted from system interventions rather than external factors such as market price fluctuations or organizational downsizing, each cost reduction was traced to a corresponding alert and incident resolution in Cherwell.

This audit trail confirmed that financial improvements originated from targeted corrective actions — such as terminating orphaned resources, enforcing policy-based shutdowns, or implementing right-sizing recommendations — all initiated through automated notifications and user responses.

Thus, the observed financial efficiency gains can be directly attributed to the implemented monitoring and escalation framework, rather than external economic conditions.

## 9. Summary

The evaluation demonstrates that the integrated monitoring solution significantly enhanced cost transparency, strengthened accountability mechanisms, and improved forecast accuracy. Furthermore, the system enabled sustained financial savings that continued to scale over time. Collectively, these results illustrate the organization's transition from a reactive cost control model to a proactive financial governance framework, aligning cost observability with both SOC 2 compliance objectives and FinOps best practices.

### 9.1. Limitations and future work

While the evaluation, grounded in the mathematical framework described in Section 4.2, demonstrates the effectiveness, several limitations must be acknowledged. First, the empirical results are derived from a single enterprise deployment, which restricts the generalizability of findings across different organizational contexts. Although aggregated monthly data for 2024 were used to enhance robustness, further validation across multiple organizations and industries would provide stronger external validity.

Second, the forecasting analysis was limited to three representative models (linear regression, random forest, and kernel regression). While these capture both interpretable and non-linear behaviors, additional methods such as gradient boosting, recurrent neural networks, or hybrid ensembles may yield further improvements in predictive accuracy.

Third, the evaluation focused on aggregated cloud spending across all providers. A more granular breakdown by cloud service (AWS, Azure, GCP) would allow for provider-specific anomaly detection and optimization strategies, thereby offering a deeper view of multi-cloud financial risk.

Future work will therefore concentrate on expanding the dataset across multiple case studies, incorporating additional forecasting techniques, and conducting provider-level analyses. These extensions would not only improve the scientific rigor of the evaluation but also strengthen the practical applicability of cost observability frameworks in SOC 2–aligned cloud environments.

## 10. Conclusions

This article examined the pressing issue of cost monitoring in a multi-cloud environment, a challenge faced by many modern IT organizations due to infrastructure scaling, diverse billing models across cloud providers, and increasing financial risks. Emphasis was placed on the need to combine technical capabilities with management practices to create a consistent and automated approach to budget tracking, cost forecasting, and timely incident response in cases of overspending.

The architectural solution analyzed in this work — which integrates the capabilities of Splunk, Cherwell, and public cloud billing APIs — demonstrates that such automation can achieve several key outcomes:

- Centralized and structured collection of billing data from AWS, Azure, and GCP.
- Timely identification of budget deviations through predefined thresholds.
- Escalation mechanisms with clear role-based responsibilities and automated notifications.
- Creation of incidents and service requests in response to budget breaches.
- Integration with broader business processes through CMDB/ITSM tools.
- Generation of visual forecasts and strategic reports via Splunk.

An evaluation of historical cost dynamics showed a tangible impact after the system was implemented: improved transparency of actual cloud spending, stabilization of total budget usage patterns, and closer alignment between approved and consumed resources. These results reflect improved financial discipline and decision-making.

## 11. Empirical results

The evaluation confirmed measurable benefits. Forecast accuracy improved through comparative model testing (linear regression, Random Forest, kernel), with average monthly errors quantified using RMSE, MAE, and $R^2$ metrics. Financial savings grew steadily: from marginal values (<$2k per month in mid-2023) to over **$34k per month by early 2025**, illustrating the scalability of proactive monitoring.

Compared to native CSP tools (e.g., AWS Cost Explorer, Azure Cost Management, GCP Billing Reports), the proposed solution provided deeper integration with ITSM processes and explicit alignment with SOC 2 criteria, leading to higher transparency, structured escalation, and stronger audit readiness.

Beyond its FinOps value, the proposed solution also contributes to compliance and audit readiness, particularly in the context of the SOC 2 framework. By embedding key controls such as role-based access, anomaly detection, escalation logging, and traceable workflows, the system addresses several TSC:

- Processing Integrity. This criterion addresses the system's ability to ensure the completeness, accuracy, and timeliness of data processing. In this case, it applies to billing and budget-related data from cloud platforms.
- The system ensures processing integrity through automated and scheduled collection of billing data from AWS, Azure, and GCP via standardized JSON APIs. The data is updated three times per day, ensuring that it remains current.
- Moreover, the collected data is systematically compared with approved budgets stored in Cherwell, allowing discrepancies to be immediately identified. This supports data consistency and reliability, which are critical for financial decision-making and compliance.
- The Security criterion focuses on protecting data from unauthorized access, modification, or loss. In the proposed model, security is maintained through Role-Based Access Control (RBAC) implemented in both Splunk and Cherwell, ensuring that users only see or act on data relevant to their responsibilities.

  Additionally, the system includes built-in anomaly detection mechanisms: when budget thresholds are exceeded, alerts are automatically triggered. In more critical cases, incident tickets are generated and escalated. This enables a proactive response to unusual or unauthorized spending behavior, reducing potential risks associated with human error or misuse.

- Availability in this case ensures that systems remain operational and accessible, even when incidents or risks arise. The solution addresses this through proactive response workflows. For example, when a project exceeds its allocated budget, the system automatically escalates the event by notifying responsible roles and creating an incident in Cherwell. These timely responses help prevent budget exhaustion, which could otherwise lead to interruption of cloud services or suspension of essential workloads.
- The Monitoring Activities criterion refers to the system's ability to track and assess the effectiveness of internal controls.

The architecture includes comprehensive event logging, capturing everything from billing data ingestion to alerts and incident creation. Logs are stored in Splunk and Cherwell, providing a detailed audit trail that supports both internal oversight and external audits.

In this way, the system not only streamlines cost management but also establishes a clear audit trail and documentation of financial decisions, which is essential for passing SOC 2 Type II assessments.

## 12. Functional value proposition

Beyond its architectural and compliance contributions, the proposed solution delivers clear enterprise value. Three categories of benefits are particularly evident:

- **Compliance automation:** By embedding cost observability into operational workflows, the system automates evidence collection and escalation logging. This reduces the manual effort required for SOC 2 audit preparation, potentially lowering audit readiness costs by up to 70 %.
- **Risk reduction:** Continuous monitoring of financial anomalies eliminates blind spots associated with shadow workloads and uncontrolled expenditures. This strengthens incident response and minimizes exposure to misconfigurations or unauthorized deployments that carry both financial and security risks.
- **Operational integration:** Financial anomalies are transformed into actionable security intelligence through SIEM/CMDB integration. This bridges FinOps and security operations, ensuring that financial risks are managed with the same rigor as technical incidents.

The financial impact observed in evaluation — with monthly cost savings scaling to **over $34,000 by early 2025** — can be directly linked to enterprise reporting. When projected as recurring operational savings, such reductions translate into significant improvements in departmental budgets and operational income. Treated as incremental cash flows, these savings could support new project funding or offset infrastructure growth over multi-year horizons, thereby reinforcing both financial discipline and business agility.

It enables IT organizations to demonstrate control, accountability, and structured risk response — all of which are critical factors in environments where trust, compliance, and data stewardship are essential to maintaining client and partner confidence. Thus, the proposed architecture functions not just as a tool for cost optimization, but as a broader governance and compliance framework that links technical events with financial accountability — supporting both operational performance and regulatory readiness.

## 13. Contribution and novelty

This work advances the state of practice by reframing cloud cost monitoring as a security control directly mapped to SOC 2 TSC (Security, Availability, Processing Integrity, Monitoring Activities). Unlike existing FinOps or CSPM approaches, the proposed system demonstrates how financial anomalies can be treated as security-relevant signals, bridging operational finance and information security.

## 14. Limitations and future work

The current evaluation is based on a single enterprise deployment.

Broader validation across organizations of different sizes and industries is required to generalize findings. Additionally, forecasting was limited to regression and ensemble models; future work should incorporate time-series methods (e.g., ARIMA, Prophet) to capture seasonal fluctuations. Finally, successful adoption presupposes robust tagging practices and integration with CMDB/ITSM platforms, which may not be feasible in less mature organizations.

## 15. Implementation guidelines for cost observability as a security control

To implement cost observability as an effective security control, organizations must align their financial monitoring processes with existing security, compliance, and operational workflows. This integration allows financial anomalies—such as unexpected cloud spending or deviations from planned budgets—to be treated not only as accounting issues but as potential indicators of misconfiguration, abuse, or systemic inefficiencies. The following guidelines outline a structured approach for implementing such a model.

First, it is essential to establish centralized data collection from all cloud providers in use. This involves integrating native APIs from platforms like AWS, Azure, and Google Cloud in a consistent, automated manner. Cost data should be updated multiple times per day to ensure near real-time visibility into consumption patterns and budget adherence.

Next, the collected data should be processed and analyzed using a robust observability platform such as Splunk. This system acts as the analytical core, responsible for aggregating data, comparing it with predefined budget thresholds, and triggering alerts. The observability platform must be tightly integrated with ITSM tools, such as Cherwell or ServiceNow, to allow for automated creation of service requests or incident records when thresholds are breached.

To make cost-related events actionable, organizations should define a tiered escalation model. For example, reaching 70 % of a defined budget may trigger a warning email to the project's budget owner, while surpassing 90 % could initiate notifications to broader teams, including DevOps and FinOps. If 100 % of the budget is exceeded and no corrective action is taken within a set period, an incident should be escalated to the security team (SecOps or DevSecOps) for further analysis.

In parallel, strict RBAC should be applied to all cost monitoring interfaces. Only authorized individuals should be allowed to access dashboards, modify thresholds, or respond to cost-related events. All actions within the system must be logged, timestamped, and preserved to support internal auditing and external compliance reviews.

Forecasting capabilities are also crucial. By using tools such as Splunk, organizations can generate dynamic reports and trend predictions, enabling proactive planning and early identification of potential overspending. These reports should be distributed on a regular basis to both technical and financial stakeholders to facilitate collaborative decision-making.

Importantly, all these practices should be mapped to the SOC 2 TSC. Cost observability systems contribute to Processing Integrity by ensuring the accuracy and timeliness of billing data; to Security by controlling access and detecting anomalies; to Availability by preventing service interruptions due to unmonitored budget exhaustion; and to Monitoring Activities through comprehensive event logging and visual analytics.

Finally, the successful deployment of such a system depends on collaboration between FinOps, DevOps, and security teams. Cost-related signals must be treated with the same level of operational urgency as other infrastructure anomalies, closing the gap between financial oversight and security posture.

In sum, implementing cost observability as a security control allows organizations not only to reduce financial risk but also to build a more resilient, transparent, and audit-ready multi-cloud infrastructure.

## Ethical approval

This research did not involve human participants, personal data, or animals.

## CRediT authorship contribution statement

**Yevhenii Martseniuk:** Writing – original draft, Resources, Methodology. **Andrii Partyka:** Validation. **Ivan Opirskyy:** Conceptualization. **Oleh Harasymchuk:** Project administration.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

No data was used for the research described in the article.

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